HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **C++17** multi-threaded services for vehicle edge computing platforms, processing approximately **1,500 data points per second**.
- Developed and maintained **CMake** build configurations and unit tests for core vehicle application components, achieving over **90% code coverage**.
- Debugged and resolved complex concurrency issues within a **Linux**-based embedded system, reducing critical bug reports by **30%**.
- Collaborated with customer engineering teams to integrate vehicle application software, participating in **requirements gathering** and **post-integration validation**.
- Authored detailed **technical documentation** for internal teams and external partners, including API specifications and system architecture diagrams.
- Wrote **Python** scripts to automate build processes and streamline data ingestion for testing environments, reducing manual effort by **40%**.
- Participated in **Agile ceremonies**, including sprint planning and technical estimation for new feature development on the vehicle application software.

SKILLS

Languages – C++, Python, Java, SQL

Tools & Technologies – Linux, CMake, Git, Jenkins, Docker, EC2, S3, Lambda, CloudWatch, Splunk, Dynatrace

Frameworks & Libraries – Spring Boot, Boost C++ Libraries, scikit-learn, pandas, NumPy

Methodologies – Agile, Scrum, CI/CD, TDD

Concepts – Multi-threading, Concurrent Programming, Data Structures, Algorithms, Edge Computing, Embedded Systems, REST APIs

PROJECTS

Vehicle Data Ingestion Service (HCL Technologies)

- Developed a high-performance **C++** data ingestion service designed to process real-time sensor data from vehicle ECUs.
- Implemented **multi-threading** and efficient data serialization to handle a throughput of **~2,000 messages per second**.
- Integrated with **Linux**-based logging mechanisms and basic **Docker** containerisation for deployment testing.

Automated Test Framework for Vehicle Applications

- Created a **Python**-based framework to automate integration testing for vehicle software modules, interacting with simulated vehicle environments.
- Wrote **unit and integration tests** for core application logic, improving defect detection rates by approximately **25%** during the testing cycle.
- Designed the framework to be extensible, allowing for the addition of new test cases and support for future vehicle features.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)